

microGPT – Mathematical Functions & Formulas

1. Token Embedding

Tokens are mapped to dense vectors using an embedding matrix.

Embedding Lookup:

$$e_i = E[\text{token}_i]$$

where $E \in \mathbb{R}^{(\text{vocab_size} \times d_{\text{model}})}$

2. Positional Encoding

Adds position information to embeddings.

Combined Representation:

$$x_i = e_i + p_i$$

where p_i is the positional embedding vector.

3. Scaled Dot Product Attention

Core operation of the Transformer.

Attention(Q, K, V):

$$\text{Attention}(Q, K, V) = \text{softmax}((QK^T) / \sqrt{d_k}) V$$

4. Softmax Function

Converts logits into probabilities.

Softmax:

$$\text{softmax}(z_i) = \exp(z_i) / \sum_j \exp(z_j)$$

5. Layer Normalization

Stabilizes training by normalizing activations.

LayerNorm:

$$\text{LN}(x) = \gamma ((x - \mu) / \sqrt{(\sigma^2 + \epsilon)}) + \beta$$

where $\mu = \text{mean}(x)$, $\sigma^2 = \text{variance}(x)$

6. Feed Forward Network (MLP)

Applies non-linear transformation.

Two-Layer MLP:

$$\text{FFN}(x) = \max(0, xW_1 + b_1) W_2 + b_2$$

7. Residual Connection

Helps gradient flow.

Residual Add:

$$y = x + F(x)$$

8. CrossEntropy Loss

Training objective for next token prediction.

Loss:

$$L = - \sum y_i \log(p_i)$$

For correct class:

$$L = - \log(p_{\text{correct}})$$

9. Gradient Descent Update

Parameter optimization step.

Update Rule:

$$\theta \leftarrow \theta - \eta \nabla L(\theta)$$

where η is learning rate.